

IoT-Based Odor Detection and Automatic Perfume Sprayer With Usage Monitoring

Project report submitted in partial fulfilment of the requirements for the course- Embedded System and Internet of Things (24ECE0202) of

Bachelor of Engineering

in

Computer Science and Engineering

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21 November, 2025

CERTIFICATE

This is to certify that the project titled “**IoT-Based Odor Detection and Automatic Perfume Sprayer With Usage Monitoring**” submitted to the **Chitkara University Institute of Engineering and Technology (CUIET)** by **Sneha (2410991692), Sidharth (2410991728), Vansham(2410991760), Vidhi(2410991775)** is a bonafide record of the work done by the students towards partial fulfilment of requirements for the course- Embedded System and Internet of Things (24ECE0202) of **Bachelor of Engineering in Computer Science and Engineering**.

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Date: 21 November 2025

Candidates' Declaration

We, **Sneha (2410991692), Sidharth (2410991728), Vansham(2410991760), Vidhi(2410991775)**, of **Group- G9(B)**, B.E. -2024 batch of Chitkara University, Punjab hereby declare that the Embedded System and Internet of Things (ES&IoT) project entitled **"IoT-Based Odor Detection and Automatic Perfume Sprayer With Usage Monitoring"** is an original work and data provided in the project report is authentic and to the best of our knowledge. This project has not been submitted by us to any other institute for the award of any other course.



Contribution Details:

Sr. No.	Student Name	Roll No.	Contact Number	Contribution in Project	Signature
1.	Sneha	2410991692	9253598954	ESP32 configuration and cloud dashboard setup.	
2.	Sidharth	2410991728	6284733589	Report writing, formatting, and documentation.	
3.	Vansham	2410991760	8708353735	Coding, debugging, and system integration.	
4.	Vidhi	2410991775	9896419009	Hardware setup and sensor interfacing	

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PROBLEM STATEMENT

Public washrooms are essential facilities used by large populations on a daily basis; however, they frequently suffer from significant hygiene-related issues that compromise user comfort, public health, and overall sanitation standards. One of the primary challenges is the accumulation of foul odors caused by ammonia, sulfur compounds, methane, and other volatile organic gases generated due to continuous usage and inadequate ventilation. Since most public washrooms rely on manual inspection, the cleaning cycles often become inconsistent, delayed, or insufficient. This traditional maintenance approach does not account for real-time variations in usage patterns, resulting in prolonged periods where washrooms remain unclean, unhygienic, and unpleasant for users. In the absence of continuous monitoring, cleaning teams are forced to respond reactively rather than proactively, which reduces the overall efficiency of sanitation management.

Furthermore, the lack of automated systems leads to an absence of meaningful data regarding footfall, odor intensity, and hygiene conditions. This prevents facility managers from optimizing manpower, maintenance schedules, consumable usage (such as air fresheners and disinfectants), and resource allocation. As a result, public washrooms often experience degraded cleanliness, increased operational costs, and poor user satisfaction. These shortcomings highlight a critical need for adopting modern technological solutions to enhance hygiene levels and improve maintenance efficiency. Therefore, the development of an **IoT-based Smart Washroom Monitoring System** is essential to enable continuous odor detection, automatic air-freshener activation, real-time user counting, and provision of actionable insights for data-driven sanitation management. Such a system would significantly elevate hygiene standards and contribute to a cleaner, safer, and smarter public environment

ABSTRACT

This project develops an IoT-based Smart Odor Detection and Automatic Air Freshening System to improve hygiene in public indoor spaces. Traditional cleaning relies on fixed schedules and manual checks, causing odors and delays. Using an MQ-135 gas sensor, the system continuously monitors harmful gases and triggers a servo motor to spray perfume when odors exceed a threshold. An IR sensor tracks users to support predictive maintenance and optimized cleaning. This low-cost, automated solution enhances hygiene, cuts manual work, and improves user experience in public washrooms and smart city setups.

Project Need

- Challenges in maintaining hygiene in public washrooms
- No real-time odor detection
- Inefficient manual cleaning cycles

Applications

- Public washrooms (malls, airports, stations)
- Offices, colleges, schools
- Hospitals and healthcare

Technology Stack

- MQ-135 Gas Sensor
- IR Sensor
- Servo Motor
- ESP32 Board
- 5V/9V Power Supply

SDGs Covered

- SDG 3 – Good Health and Well-Being
- SDG 6 – Clean Water and Sanitation
- SDG 11 – Sustainable Cities and Communities

LIST OF FIGURES

Figure No.	Figure Title	Page No.
Fig 3.1	Working of MQ-135 Sensor	15
Fig 3.2	Circuit Diagram	16
Fig 3.2	Flow Chart	18
Fig 4.1	Adafruit IO dashboard	19
Fig 4.2	Project in different angles	22

TABLE OF CONTENTS

	Page No
Certificate	1
Candidates' Declaration	
Acknowledgement	3
Problem statement	4
Abstract	5
List of Figures	6
Table of Contents	7
1 INTRODUCTION	8
2 PROJECT WORK	10
3 HARDWARE & SOFTWARE	13
4 RESULTS	19
5 CONCLUSION AND FUTURE SCOPE	24
 References	
 Appendix A	
 Source Code	

CHAPTER 1

INTRODUCTION

1.1 Background of the Problem

Public washrooms are essential facilities that support daily life in schools, offices, malls, hospitals, and transportation hubs. However, they are frequently associated with foul odors, poor cleanliness, and discomfort due to high user traffic and the lack of real-time maintenance. Odorous gases such as ammonia, methane, and volatile organic compounds (VOCs) accumulate rapidly in enclosed sanitary spaces, especially when ventilation is poor and cleaning does not follow actual usage patterns. Since most washroom maintenance relies on periodic manual inspections, sudden spikes in odor or unexpected increases in user load often go undetected. This delay results in unhygienic conditions, unpleasant user experiences, and potential health concerns. With growing urban populations and increased public infrastructure usage, maintaining sanitary conditions through conventional methods has become increasingly challenging. This highlights the urgent need for automated, intelligent systems that can continuously monitor environmental conditions and respond immediately when hygiene levels begin to decline.

1.2 Need for an Accessible and Portable Assistive Solution

Manual inspection alone is insufficient for ensuring consistent hygiene because it is time-consuming, labor-intensive, and heavily dependent on human availability. Fixed cleaning schedules do not reflect real-time washroom conditions and often lead to either over-cleaning or delayed cleaning. In this context, IoT-based systems offer a practical and efficient alternative by enabling continuous, sensor-driven monitoring of odor levels, user frequency, and environmental quality. Automated odor detection systems not only provide timely identification of foul smells but also ensure immediate corrective action through automated mechanisms. By reducing the dependency on manual supervision, such systems improve hygiene standards, enhance operational efficiency, and support better maintenance planning. Therefore, an IoT-enabled solution becomes essential for maintaining consistently clean and healthy washroom environments in high-traffic public spaces.

1.3 Proposed Solution – Project Overview

The proposed system integrates multiple sensors and automated components to create a smart hygiene management solution. An MQ-135 gas sensor continuously monitors air quality and detects the presence of harmful or unpleasant gases. An IR sensor counts user entries to track washroom usage patterns, enabling data-driven cleaning schedules. When the odor level crosses a predefined threshold, the microcontroller processes the sensor inputs and triggers a servo motor to spray air freshener automatically. This ensures immediate deodorization without requiring human intervention. Designed to be compact, economical, and easily deployable, the system operates autonomously and provides a reliable way to maintain a fresh and hygienic washroom environment at all times

1.4 Scope of the Project

The scope of this project extends across a wide range of public and private environments where hygiene and air quality are critical. The system can be deployed in public washrooms located in malls, airports, railway stations, bus terminals, and educational institutions. It is equally suitable for offices, hotels, hospitals, and other high-traffic indoor facilities. Beyond odor control, the system's usage data can assist building management teams in optimizing cleaning schedules and resource allocation. The project also contributes to the broader development of smart infrastructure by integrating IoT-based automation into sanitation systems. With the potential for future enhancements such as mobile monitoring and cloud connectivity, the system can evolve into a comprehensive hygiene management platform aligned with smart city initiatives.

CHAPTER 2

PROJECT WORK

2.1 Problem Identified

Public washrooms experience rapid odor buildup due to high user traffic, poor ventilation, and the absence of real-time monitoring, resulting in unpleasant and unhygienic environments. Traditional maintenance systems rely heavily on periodic manual inspection, which fails to detect sudden spikes in odor or increased usage during peak hours. As a result, users frequently encounter foul smells, discomfort, and inconsistent levels of cleanliness. Manual odor control methods such as scheduled spraying or visual inspection are often ineffective, labor-intensive, and unable to maintain hygiene standards across different facilities. This creates the need for an automated, intelligent solution capable of detecting odors as they occur, responding instantly, and reducing dependence on human supervision.

2.2 Proposed Solution and System Development

To address these challenges, the project proposes an IoT-based Smart Odor Detection and Automatic Air Freshening System designed to operate autonomously and maintain hygiene based on real-time air quality data. The system utilizes an MQ-135 gas sensor to monitor odor-causing gases such as ammonia and VOCs, enabling precise detection of foul smells. An IR sensor records user entries, providing additional insights into washroom usage patterns that assist in better planning of cleaning schedules. When odor levels exceed a predefined threshold, a servo motor triggers a standard perfume spray, ensuring timely deodorization without requiring manual intervention. A microcontroller serves as the central processing unit, integrating sensor data, controlling the servo mechanism, and maintaining continuous operation. The system is developed using a cost-effective and modular design approach, making it suitable for deployment in various public and private facilities.

2.3 Working of the Project

The system operates by continuously reading air quality data from the MQ-135 gas sensor, which outputs analog values corresponding to the concentration of odor-causing gases. These readings are compared with a predefined threshold value programmed into the microcontroller. If the odor concentration exceeds the threshold, the microcontroller activates the servo motor to press the air freshener spray and release a burst of fragrance into the environment. Meanwhile, the IR sensor positioned at the entrance detects user movement and increments a usage counter, allowing the system to monitor washroom traffic patterns. All components work together seamlessly to provide real-time odor management, ensuring that the washroom remains clean and fresh throughout the day. The automated operation eliminates the need for continuous human monitoring, enhancing efficiency and consistency in hygiene maintenance.

2.4 Uniqueness of the Project

This project stands out due to its integration of real-time odor monitoring, automated air freshening, and user tracking within a single low-cost solution tailored specifically for public hygiene environments. Unlike conventional fragrance dispensers that operate on fixed intervals, this system functions only when required, ensuring efficient use of consumables and reducing unnecessary spraying. The addition of an IR-based usage counter provides valuable data that can significantly improve cleaning schedules and facility management. The project's modular, easily deployable design allows it to be implemented in various washroom settings without requiring expensive or proprietary equipment. By combining automation, affordability, and practicality, the system offers a unique and scalable hygiene-management solution.

2.5 Challenges Faced During Development

Several challenges were encountered during the development of this system. Calibrating the MQ-135 gas sensor was difficult due to its sensitivity to humidity, temperature, and ventilation conditions, which affected accuracy and required extensive testing to determine a reliable odor threshold. The mechanical design of the servo-based spray mechanism also required fine adjustments to ensure consistent pressure and proper actuation. Integrating the IR sensor to accurately detect user movement without false triggers was another technical challenge, particularly in

varying lighting conditions. Maintaining stable power delivery for continuous operation while avoiding interference across components also required troubleshooting. Overcoming these challenges helped refine the overall reliability and functionality of the system.

2.6 Advantages and Outcomes of the Project

The developed system offers a practical and efficient hygiene-management solution by combining real-time odor detection with automated air freshening, ensuring that foul smells are addressed immediately and consistently. Its sensor-driven operation prevents excessive use of consumables, reducing costs and conserving resources. The inclusion of a user-counting mechanism provides valuable data for optimizing cleaning schedules and improving washroom maintenance strategies. During prototype testing, the sensor module demonstrated accurate detection of odor changes, and the servo mechanism responded reliably, confirming the system's effectiveness. These features make the device highly suitable for deployment in malls, offices, colleges, hospitals, and transportation hubs where maintaining consistent hygiene is essential.

2.7 Shortcomings and Future Improvements

Despite its effective performance, the system has certain limitations, including the MQ-135 sensor's sensitivity to environmental variations such as humidity and ventilation, which may affect detection accuracy in larger or open washrooms. The threshold-based odor evaluation also limits its adaptability to rapidly shifting air quality conditions. Currently operating as a standalone model, the system lacks cloud connectivity for remote monitoring or data visualization. Future enhancements may include integrating Wi-Fi or IoT cloud platforms for real-time dashboards, implementing machine learning algorithms for predictive odor analysis, and developing a more robust dispensing mechanism to increase durability. Additional sensors and mobile app support can further improve accuracy, scalability, and user convenience.

CHAPTER 3

HARDWARE & SOFTWARE

3.1 Introduction

The Smart Washroom Monitoring System presented in this project combines multiple embedded hardware components with powerful IoT-based software tools to automate hygiene management in public and commercial washrooms. The system is designed to continuously monitor odor levels, track user frequency, and automatically trigger the air-freshening mechanism whenever necessary. With the integration of sensors, actuators, cloud dashboards, and real-time communication protocols, the project demonstrates how IoT improves sanitation, reduces manual effort, and enhances user experience in shared spaces.

The ESP32 microcontroller serves as the core processing platform, collecting data from the MQ-135 gas sensor and IR people counter sensor. When odor concentration crosses a specified threshold, the ESP32 activates a servo motor that simulates pressing an air-freshener spray. At the same time, all events—including odor percentage, spray status, and people count—are sent to the Adafruit IO dashboard using the MQTT protocol. This ensures remote monitoring and data logging for maintenance teams. The chapter explains each hardware component in detail, including their working principles, pin diagrams, interfacing methods, the block diagram flow, circuit connections, software tools, libraries used, and operational workflow of the system.

3.2 Hardware Components Used

- The project uses an ESP32 microcontroller, which serves as the main processing unit responsible for reading sensor data, calculating odor percentage, controlling the servo motor, and sending data to Adafruit IO via WiFi.
- The MQ-135 gas sensor detects washroom odors and harmful gases using its SnO₂ sensing layer, and its analog output connected to GPIO 34 increases with gas concentration, triggering the spray mechanism when odor exceeds 50%.
- An IR sensor module connected to GPIO 27 counts users by detecting IR beam interruptions, helping generate alerts when usage reaches predefined thresholds such as 5 (cleaning required) and 10 (counter reset).

- The SG90 servo motor, controlled through PWM on GPIO 13, simulates pressing an air freshener bottle by rotating between 10° and 180° during high-odor events.
- Additional components like a breadboard, jumper wires, 5V power supply, and mounting supports ensure stable circuit connections and proper physical installation of the system.

3.3 ESP32 Pin Configuration

The ESP32 board contains 30–38 pins depending on the version. The important pins used in the project include:

Pin No.	Pin Name	Function
GPIO 34	ADC Input	Reads MQ-135 analog output
GPIO 27	Digital Input	Reads IR sensor signal
GPIO 13	PWM Output	Controls servo motor
5V Pin	Power	Power sensors and servo
GND	Ground	Common ground reference

Table 3.1: ESP32 Pin Configuration

3.4 Working of MQ-135 Sensor

After a warm-up time of approximately 20 seconds, the sensor begins detecting gases. The analog voltage is read by the ESP32, which maps the value to a scale of 0–100%.

Odor Computation Flow:

1. Read raw analog value
2. Map range 700–3000 to 0–100%
3. Compare with threshold (50%)
4. If above 50% → trigger spray
5. Publish sensor status to cloud

The sensor is sensitive to humidity, temperature, and surrounding chemicals, making it ideal for washrooms where odor sources vary.



Figure 3.1: Working of MQ-135 Sensor

3.5 Circuit Description

The circuit is simple and modular, allowing easy installation in real environments.

- MQ-135 connects to the ADC input of ESP32.
- IR sensor connects to digital input.
- Servo motor connects to a PWM-enabled GPIO.
- All components share a common ground.
- WiFi connection is used for cloud communication.

Power is provided through a 5V adapter or a USB cable connected to the ESP32. The servo motor draws more current during rotation, so proper grounding ensures stable operation.

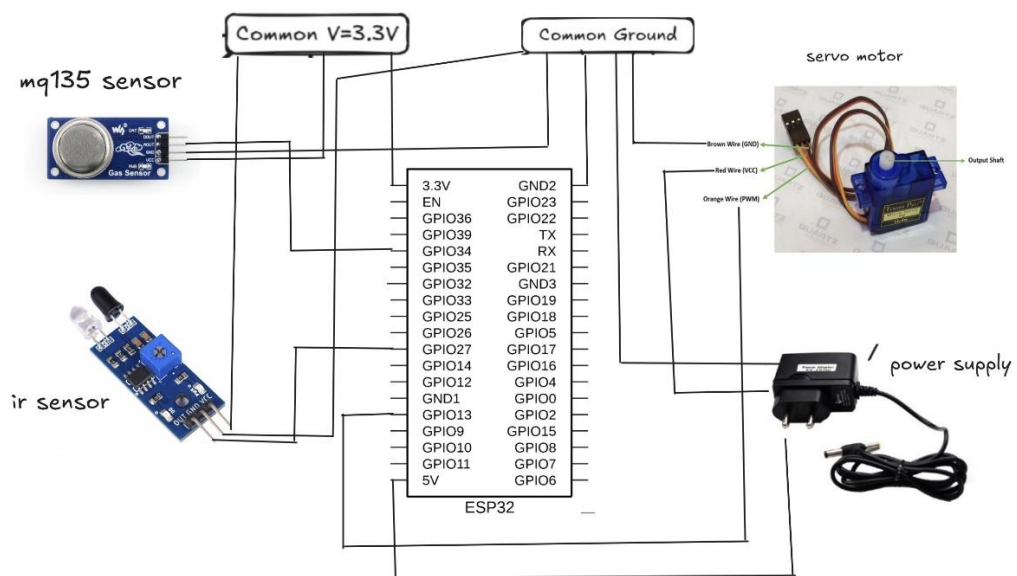


Figure 3.2: Circuit Diagram

3.6 Software Tools

(a) Arduino IDE

Used to write, compile, and upload code to the ESP32. Includes built-in Serial Monitor for debugging.

(b) Adafruit IO Cloud Dashboard

A web-based IoT dashboard used to visualize:

- Odor percentage
- People count
- Cleaning alerts
- Spray status

(c) MQTT Protocol

Lightweight communication protocol used for publishing data to the cloud efficiently.

(d) Serial Debugging Tools

Used to view sensor values, WiFi status, and real-time system messages.

3.7 Libraries Used

The following libraries were essential in implementing system functionality:

Library Name	Purpose
WiFi.h	Connects ESP32 to WiFi
Adafruit_MQTT.h	Manages MQTT connection
Adafruit_MQTT_Client.h	Publishes IoT data to Adafruit IO
ESP32Servo.h	Controls servo motor angles
Default Arduino Core	Handles GPIO, ADC, timing functions

Table 3.2: Libraries Used

3.8 Flowchart of System Operation

Start → Initialize ESP32 → Connect WiFi → Warm MQ-135 → Read Odor → If Odor > 50% → Spray → Count People → Check Threshold → Send Alerts → Update Cloud → Loop Again

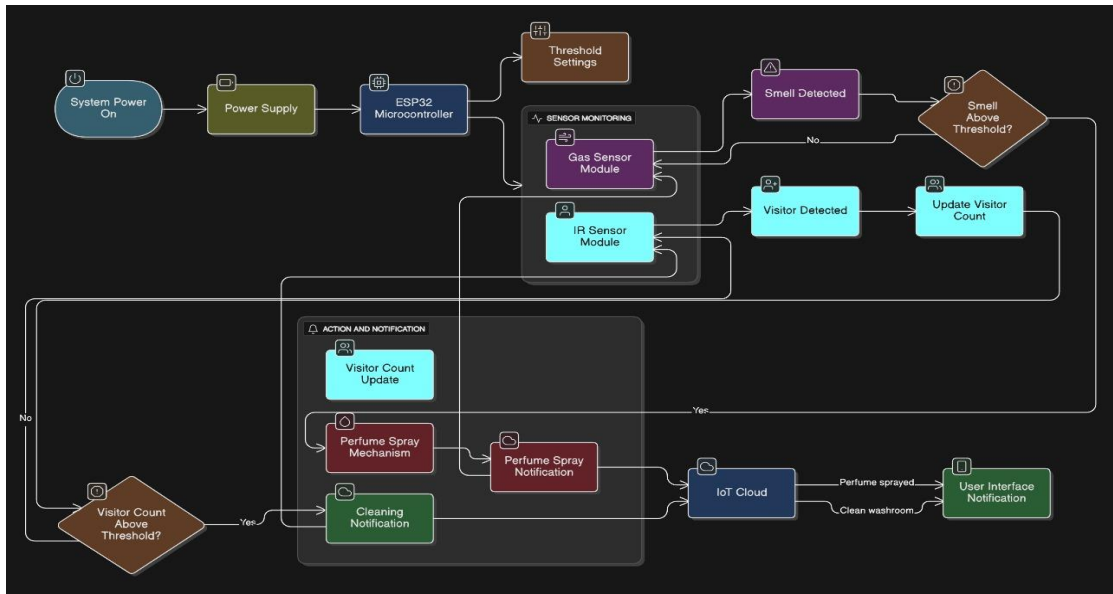


Figure 3.3: FlowChart

3.9 Summary of Chapter

This chapter presented a detailed explanation of the hardware and software architecture used in the Smart Washroom Monitoring System. It included the working principles, pin configuration, block and circuit diagrams, and complete workflow of odor detection, people counting, cloud integration, and servo activation. The combination of sensors, microcontroller, IoT cloud, and automation creates a highly efficient and scalable hygiene-management system suitable for public washrooms, offices, malls, and institutional buildings.

CHAPTER 4

RESULTS

4.1 System Performance and Output

The IoT-based Smart Washroom Monitoring System showed reliable performance by accurately detecting odor levels, counting users, and triggering the spray mechanism whenever required. The MQ-135 sensor provided stable readings that the ESP32 converted into odor percentage values, updated every 10 seconds for real-time monitoring. When the odor level exceeded 50%, the servo motor activated smoothly for 5 seconds to simulate an air-freshener spray.

The IR sensor correctly registered user entries using its HIGH-to-LOW detection method, enabling accurate counting. Cleaning alerts were generated at 5 users, and an automatic reset occurred at 10 users. The system also performed well on Adafruit IO, where odor values, spray status, and people count updated with minimal delay. MQTT communication remained stable even under varying WiFi conditions. Overall, the system functioned efficiently, providing quick and automated hygiene management.

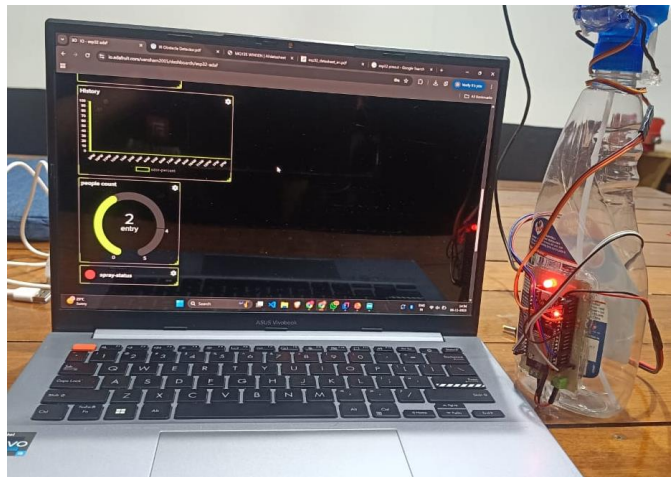


Figure 4.1 Adafruit IO dashboard

4.2 Testing and Evaluation

Extensive testing was performed to evaluate system reliability, accuracy, and performance under different environmental conditions. The MQ-135 sensor was tested across varying odor intensities, including sprays, deodorants, and natural washroom odors. Calibration proved accurate, with odor percentages increasing proportionally to the concentration of gases. The servo motor's motion sequence was verified at least 20 times, and the spraying mechanism consistently completed its 5-second sweep without overheating or stalling.

The IR sensor was tested by simulating multiple people entering the washroom. It accurately detected transitions and maintained count precision without false triggers. Different lighting conditions were tested, and the IR sensor performed reliably in all scenarios. Network testing confirmed that MQTT data packets were delivered to Adafruit IO with negligible delay, and the dashboard updated all fields nearly instantaneously.

A real-time test run for 30 minutes simulated washroom usage, odor rise, spray activation, and cloud logging. All system functionalities worked as expected, validating the practical usability of the prototype.

4.3 Application Areas of the Project

The Smart Washroom Monitoring System has a wide range of applications across public and private infrastructure where hygiene and odor control are essential. In **public washrooms** at malls, metro stations, airports, and bus stands, the system can drastically reduce complaints related to bad odors by ensuring timely spray activation and usage-based cleaning alerts. In **corporate offices, colleges, and schools**, the system offers a cost-effective way to maintain clean washrooms without requiring manual inspection.

In **hospitals and healthcare centers**, where sanitation is critical, the system helps in maintaining odor-free and hygienic conditions for patients and staff. Hotels, restaurants, and cafés can use the system to improve customer satisfaction by ensuring consistently fresh restrooms. The system is also beneficial in **smart buildings and smart city initiatives**, supporting autonomous sanitation management. Industries dealing with mild gases or chemical odors can use the system for air-quality improvement inside restrooms or small workspaces.

4.4 Comparison with Existing Solutions

Existing washroom hygiene systems rely heavily on manual cleaning cycles and human inspection. Traditional air fresheners work on fixed timers, which often lead to over-spraying or under-spraying, creating inefficiency and wastage. In contrast, this IoT-based system activates spraying **only when odor levels actually rise**, making it more intelligent and resource-efficient.

Compared to standalone air-quality sensors that only display readings but do not take action, this system provides a **complete automated solution** by combining odor detection with servo-based actuation and cloud-based alerts. Unlike high-cost commercial sanitation systems, this project uses affordable components and an open-source IoT platform, making it suitable for large-scale deployment. The use of Adafruit IO for cloud monitoring provides visibility from anywhere, unlike conventional fresheners that lack monitoring features. Overall, the proposed system is more economical, smarter, and significantly more functional than existing alternatives.

4.5 Advantages of the System

The Smart Washroom Monitoring System provides several advantages that make it suitable for real-world implementation.

- It operates automatically, reducing dependence on human supervisors.
- Real-time odor detection ensures immediate air freshening and improved user experience.
- People counting helps facility staff clean washrooms based on actual usage rather than fixed schedules.
- Integration with Adafruit IO enables remote monitoring with clear visual dashboards.
- The system minimizes consumable wastage by triggering sprays only when needed.
- Compact and low-cost hardware makes it scalable for multiple washrooms.
- The ESP32 ensures fast processing, low power consumption, and stable cloud connectivity.

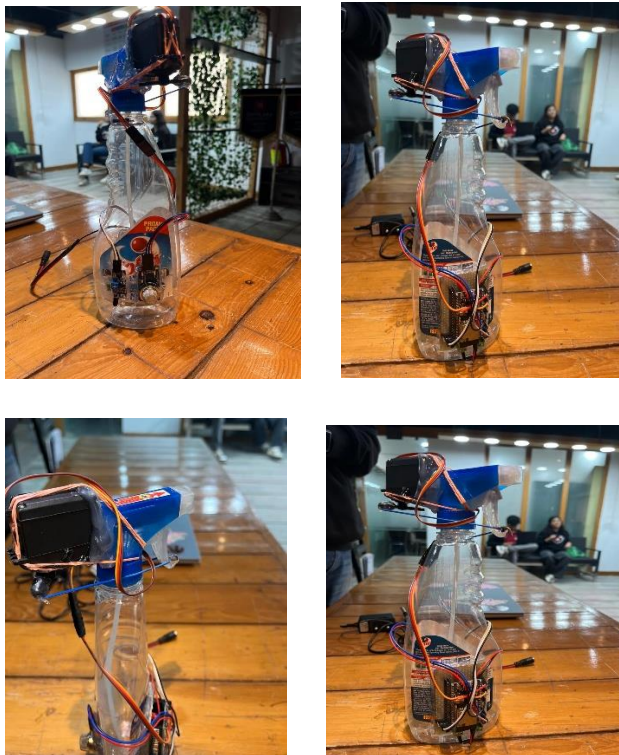


Figure 4.2 Project in different angles

4.6 Limitations of the System

Despite successful working, the system has certain limitations. The MQ-135 sensor can be influenced by humidity and ventilation, which may slightly affect accuracy in very large or open washrooms. The IR sensor may sometimes miss counts if multiple people pass simultaneously or too quickly. The system currently uses a simple threshold-based decision mechanism; more advanced AI-based odor prediction could improve intelligence. The servo motor mechanism works well for demonstration but may need a stronger mechanism for long-term commercial use. The system also requires stable WiFi for continuous cloud updates, although local automation continues even without internet.

4.7 Overall System Outcome

The final prototype successfully fulfilled its objective of creating an automated, IoT-enabled smart washroom monitoring solution. The system effectively detected odor changes, counted users, generated cleaning alerts, handled automated spray activation, and uploaded all data to Adafruit IO in real time. The project demonstrated that even with low-cost sensors and compact hardware, an efficient hygiene-management solution can be developed.

The results confirm that the system is practical, scalable, and suitable for deployment in public washrooms, institutions, corporate offices, and smart city infrastructures. The performance evaluation indicates strong potential for future expansion, including ML-based odor prediction, multi-sensor fusion, advanced data analytics, and fully automated smart sanitation ecosystems.

CHAPTER 5

Conclusion and Future Scope

5.1 Conclusion

The project **“IoT-Based Smart Washroom Monitoring System using ESP32”** successfully demonstrates an automated and intelligent hygiene-management solution capable of detecting odor levels, monitoring washroom usage, and activating an air-freshening mechanism without human intervention. By integrating the MQ-135 air-quality sensor, IR people counter, servo motor, and cloud connectivity through Adafruit IO, the system ensures real-time odor monitoring and timely hygiene maintenance. The ESP32 microcontroller efficiently processed sensor data, triggered spray actions, and published all events to the IoT platform, offering complete transparency and remote supervision.

Through testing, the system proved to be reliable, responsive, and suitable for deployment in public washrooms, offices, hospitals, institutions, transportation hubs, and smart city environments. The automated functionality reduces manual supervision, improves user experience, and supports data-driven cleaning schedules. Overall, the project achieves its intended objective of providing a low-cost, scalable, and practical IoT-based sanitation solution for maintaining clean and healthy washroom conditions.

5.2 Learnings from the Project

During the development of this project, the team gained extensive hands-on experience with embedded systems, electronics, and IoT integration. Key learnings included understanding ESP32 architecture, ADC and PWM functionalities, interfacing sensors such as MQ-135 and IR modules, and working with servo motors for actuation. The team learned how to implement real-time data acquisition, threshold-based automation, and cloud-based data publishing using MQTT.

Important practical skills gained include debugging hardware circuits, calibrating sensors, analyzing sensor outputs, optimizing timing intervals, and handling WiFi-based communication. The project also strengthened knowledge in system design, teamwork, documentation, and problem-solving—skills essential for engineering development.

5.3 Technologies and Software Learned

Throughout the project, students worked with a variety of technologies, tools, and libraries, including:

- Arduino IDE for ESP32 programming
- ESP32 WiFi libraries for wireless communication
- MQTT protocol for efficient cloud data transfer
- Adafruit IO dashboard for IoT visualization and monitoring
- ESP32Servo library for controlling the servo motor
- Serial Monitor for real-time debugging and testing
- IoT networking concepts such as client-server communication, feed publishing, and dashboard creation.

5.4 Benefits to Students and Society

This project helps students gain practical skills in embedded systems, IoT communication, cloud dashboards, and real-time automation. It strengthens hands-on problem-solving, sensor calibration, coding, and debugging abilities, making students more industry-ready.

For society, the system provides an effective solution to unhygienic washrooms by automating odor monitoring and cleaning alerts. This reduces manual supervision, improves cleanliness, prevents foul-smell-related discomfort, and promotes better hygiene standards in public spaces. It also supports smart-city digital sanitation initiatives by enabling data-driven maintenance

5.5 Future Scope / Improvements

Future enhancements can include adding sensors like DHT22 or CO₂ modules for improved air-quality analysis and using machine learning for predicting odor levels. Mobile app integration may allow direct alerts to staff. The spray mechanism can be upgraded to a solenoid-based automatic dispenser for commercial durability. Additional features like automatic exhaust fan control, maintenance history logs, and advanced dashboards can make the system more robust and suitable for large-scale deployment.

5.6 Shortcomings of the Project

The MQ-135 sensor is influenced by temperature and humidity, which can slightly affect accuracy. The IR sensor may miscount when multiple users pass simultaneously. The prototype spray setup requires a more durable design for long-term use. Cloud updates depend on WiFi stability, and odor thresholds may need recalibration in different environments. These limitations can be improved in the next version for better reliability and performance.

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(Used for IR sensor working and digital output behavior)

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(Used for servo control signals, angle range, and PWM mechanism)

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(Used for understanding publish/subscribe MQTT feeds and cloud dashboard operations)

[6] S. Gupta and M. Kaur, “IoT-Based Smart Washroom and Air Quality Monitoring System,” *International Journal of Computer Applications*, vol. 182, no. 29, 2019.

(Related research work supporting smart sanitation systems)

APPENDIX A

Source Code of the Project :

```
#include <WiFi.h>
#include "Adafruit_MQTT.h"
#include "Adafruit_MQTT_Client.h"
#include <ESP32Servo.h>

//***** Wi-Fi & Adafruit IO Configuration *****/
#define WIFI_SSID "Apna Net Use Kro"
#define WIFI_PASS "mathematics"
#define AIO_SERVER "io.adafruit.com"
#define AIO_SERVERPORT 1883
#define AIO_USERNAME "vansham2005"
#define AIO_KEY "aio_OvIP45qwdkVPXsv8NwBctIO2meVm"

//***** Pin Definitions *****/
const int MQ_SENSOR_PIN = 34; // ADC Pin for MQ-135
#define IR_SENSOR_PIN 27 // IR people counter
#define SERVO_PIN 13 // Servo motor (spray mechanism)

//***** Sensor Calibration & Thresholds *****/
const int MIN_SENSOR_VALUE = 700; // Clean air (0%)
const int MAX_SENSOR_VALUE = 3000; // Polluted air (100%)
#define ODOR_THRESHOLD 50 // Trigger spray at 50%
#define CLEANING_THRESHOLD 5 // Alert after 5 people
#define RESET_THRESHOLD 10 // Reset counter at 10 people

//***** Motor/Spray Control Settings *****/
#define SPRAY_DURATION 5000 // Motor runs for 5 seconds

//***** Global Variables *****/
// Networking
WiFiClient client;
Adafruit_MQTT_Client mqtt(&client, AIO_SERVER, AIO_SERVERPORT, AIO_USERNAME, AIO_KEY);

// Adafruit IO Feeds
void Adafruit_MQTT_Publish odorPercentFeed = Adafruit_MQTT_Publish(&mqtt,
    AIO_USERNAME "/feeds/odor-percent");
void Adafruit_MQTT_Publish odorStatusFeed = Adafruit_MQTT_Publish(&mqtt,
    AIO_USERNAME "/feeds/odor-status");
void Adafruit_MQTT_Publish peopleCountFeed = Adafruit_MQTT_Publish(&mqtt,
    AIO_USERNAME "/feeds/people-count");
void Adafruit_MQTT_Publish sprayStatusFeed = Adafruit_MQTT_Publish(&mqtt,
    AIO_USERNAME "/feeds/spray-status");
void Adafruit_MQTT_Publish cleaningAlertFeed = Adafruit_MQTT_Publish(&mqtt,
    AIO_USERNAME "/feeds/cleaning-alert");

// IR Counter State
int peopleCount = 1; // Start from 1
int lastIrStatus = HIGH;
bool cleaningAlertActive = false;

// Servo Motor State
Servo myservo;
int servoPos = 90; // Start at center position
bool motorRunning = false;
unsigned long motorStartTime = 0;

// Odor Reading Timer
unsigned long lastOdorReadingTime = 0;
const long odorReadingInterval = 10000; // Check every 10 seconds

// Function Prototypes
void MQTT_connect();
void activateSprayMotor();
void stopSprayMotor();
void updateMotor();
```

```

void setup() {
  Serial.begin(115200);
  delay(10);

  Serial.println(F("\n"));
  Serial.println(F("IoT Smart Washroom Monitoring System - VIVA DEMO"));
  Serial.println(F("\n"));

  // Initialize Pins
  pinMode(IR_SENSOR_PIN, INPUT);
  myservo.attach(SERVO_PIN);
  myservo.write(90); // Center position (motor idle)
  Serial.println(F("✅ Servo motor initialized at 90° (IDLE position)"));

  // Connect to WiFi
  Serial.print(F("\n⚙️ Connecting to WiFi: "));
  Serial.print(WIFI_SSID);
  Serial.print(F(" "));
  WiFi.begin(WIFI_SSID, WIFI_PASS);

  int wifiAttempts = 0;
  while (WiFi.status() != WL_CONNECTED && wifiAttempts < 20) {
    delay(500);
    Serial.print(".");
    wifiAttempts++;
  }

  if (WiFi.status() == WL_CONNECTED) {
    Serial.println(F("\n✅ WiFi Connected Successfully!"));
    Serial.print(F("📍 IP Address: "));
    Serial.println(WiFi.localIP());
  }
}

void loop() {
  MQTT_connect();
  unsigned long currentMillis = millis();

  //-----
  // SECTION A: MQ-135 ODOR DETECTION & MOTOR CONTROL
  //-----

  if (currentMillis - lastOdorReadingTime >= odorReadingInterval) {
    lastOdorReadingTime = currentMillis;

    Serial.println(F("\n-----"));
    Serial.println(F("👉 CHECKING AIR QUALITY..."));
    Serial.println(F("-----"));

    // Read sensor
    int sensorValue = analogRead(MQ_SENSOR_PIN);
    long odorPercent = map(sensorValue, MIN_SENSOR_VALUE, MAX_SENSOR_VALUE, 0, 100);
    odorPercent = constrain(odorPercent, 0, 100);

    Serial.print(F("📊 Raw Sensor Value: "));
    Serial.println(sensorValue);
    Serial.print(F("📊 Odor Percentage: "));
    Serial.print(odorPercent);
    Serial.println(F("%"));

    // Publish to Adafruit IO
    if (odorPercentFeed.publish((int32_t)odorPercent)) {
      Serial.println(F("✅ Odor percentage sent to Adafruit IO"));
    } else {
      Serial.println(F("❌ Failed to send odor percentage"));
    }
  }
}

```

```

if (odorPercent > ODOR_THRESHOLD) {
  Serial.println(F("\n 🚨 HIGH ODOR DETECTED!"));
  Serial.print(F("   Odor Level: "));
  Serial.print(odorPercent);
  Serial.println(F("% (Above 50% threshold)"));

  odorStatusFeed.publish("high odor");
  Serial.println(F("   ✅ Status 'high odor' sent to Adafruit IO"));

  // Activate spray motor
  Serial.println(F("\n 🚿 ACTIVATING SPRAY MOTOR..."));
  activateSprayMotor();
} else {
  Serial.println(F("\n ✅ AIR QUALITY NORMAL"));
  Serial.print(F("   Odor Level: "));
  Serial.print(odorPercent);
  Serial.println(F("% (Below 50% threshold)"));

  odorStatusFeed.publish("low odor");
  Serial.println(F("   ✅ Status 'low odor' sent to Adafruit IO"));
  Serial.println(F("   🛑 Motor remains IDLE (no spray needed)"));
}

Serial.println(F("-----\n"));

```

```

    cleaningAlertActive = true;
  } else {
    Serial.println(F("   ❌ Failed to send cleaning alert"));
  }

  Serial.println(F("\n   🛑 Alert will persist until count reaches 10"));
} else if (peopleCount < CLEANING_THRESHOLD) {
  // Before cleaning threshold
  Serial.print(F("   🛑 "));
  Serial.print(CLEANING_THRESHOLD - peopleCount);
  Serial.println(F(" more people needed for cleaning alert"));
} else if (peopleCount > CLEANING_THRESHOLD && peopleCount < RESET_THRESHOLD) {
  // Between 5 and 10
  Serial.print(F("   🚨 CLEANING REQUIRED! ("));
  Serial.print(RESET_THRESHOLD - peopleCount);
  Serial.println(F(" more people until counter reset)"));
}

// Check if reset threshold reached (EXACTLY at 10)
if (peopleCount == RESET_THRESHOLD) {
  Serial.println(F("\n 🔄 COUNTER RESET TRIGGERED! 🔄"));
  Serial.println(F("   🔄 COUNTER RESET TRIGGERED! 🔄"));
  Serial.println(F(""));
  Serial.println(F("   🔄 10 people threshold reached"));
  Serial.println(F("   🧹 Assuming washroom has been cleaned"));
  Serial.println(F("   🔄 Resetting counter to 1..."));

  // Reset counter to 1
  peopleCount = 1;
  cleaningAlertActive = false;
}

```

```

    // Publish reset count
    if (peopleCountFeed.publish((int32_t)peopleCount)) {
        Serial.println(F("   ✅ Counter reset to 1 sent to Adafruit IO"));
    }

    // Clear cleaning alert (send empty message)
    if (cleaningAlertFeed.publish("")) {
        Serial.println(F("   ✅ Cleaning alert CLEARED on Adafruit IO"));
    }

    Serial.println(F("\n   ✅ System reset complete!"));
    Serial.println(F("   ✅ Ready to count from 1 again"));
    Serial.println(F("   ✅ Cleaning message removed from dashboard"));
}

Serial.println(F("-----\n"));

delay(500); // Debounce
}

lastIrStatus = currentIrStatus;

//=====
// SECTION C: MOTOR UPDATE (Runs during spray sequence)
//=====

updateMotor();

// Keep MQTT alive
mqtt.processPackets(100);
delay(15); // Small delay for stability
}

int currentIrStatus = digitalRead(IR_SENSOR_PIN);

// Edge detection: HIGH to LOW transition
if (currentIrStatus == LOW && lastIrStatus == HIGH) {
    peopleCount++;

    Serial.println(F("\n-----"));
    Serial.println(F(" 🚶 PERSON DETECTED!"));
    Serial.println(F("-----"));
    Serial.print(F(" 🧑 Total People Count: "));
    Serial.print(peopleCount);
    Serial.print(F(" / "));
    Serial.println(RESET_THRESHOLD);

    // Publish count
    if (peopleCountFeed.publish((int32_t)peopleCount)) {
        Serial.println(F("   ✅ Count sent to Adafruit IO"));
    } else {
        Serial.println(F("   ❌ Failed to send count"));
    }

    // Check if cleaning threshold reached (EXACTLY at 5)
    if (peopleCount == CLEANING_THRESHOLD) {
        Serial.println(F("\n-----"));
        Serial.println(F(" 🚧   🚧   CLEANING ALERT TRIGGERED!   🚧   🚧   "));
        Serial.println(F("-----"));
        Serial.println(F(""));
        Serial.println(F(" 🛠️ CLEANING REQUIRED!"));
        Serial.println(F(" 🚦 Threshold of 5 people reached"));
        Serial.println(F(" 📡 Sending alert to Adafruit IO dashboard..."));

        // Send cleaning alert
        if (cleaningAlertFeed.publish("CLEANING REQUIRED")) {
            Serial.println(F("   ✅ Cleaning alert sent successfully!"));
        }
    }
}

```

```

void activateSprayMotor() {
    motorRunning = true;
    motorStartTime = millis();
    servoPos = 10; // Start from left position

    Serial.println(F(" 🚧 Motor starting sweep sequence..."));
    Serial.println(F(" 🚧 Motor will sweep 10° → 180° for 5 seconds"));

    // Publish spray status as SPRAYING (will show RED on dashboard)
    if (sprayStatusFeed.publish("SPRAYING")) {
        Serial.println(F(" ✅ Spray status 'SPRAYING' (RED) sent to Adafruit IO"));
    }

    Serial.println(F("\n 🚧 SPRAYING IN PROGRESS 🚧"));
    Serial.println(F(" 🚧 SPRAYING IN PROGRESS 🚧"));
    Serial.println(F(" 🚧 SPRAYING IN PROGRESS 🚧"));
}

/*****
 * STOP SPRAY MOTOR
 * Stops the motor and returns to idle position
 *****/
void stopSprayMotor() {
    motorRunning = false;
    myservo.write(90); // Return to center (idle position)

    Serial.println(F("\n ✅ SPRAYING COMPLETE ✅"));
    Serial.println(F(" ✅ SPRAYING COMPLETE ✅"));
    Serial.println(F(" ✅ SPRAYING COMPLETE ✅"));
    Serial.println(F(" 🏠 Motor returned to IDLE position (90°)"));

    // Publish spray status as IDLE (will show GREEN on dashboard)
    if (sprayStatusFeed.publish("IDLE")) {
        Serial.println(F(" ✅ Spray status 'IDLE' (GREEN) sent to Adafruit IO\n"));
    }
}

/*****
void MQTT_connect() {
    int8_t ret;

    if (mqtt.connected()) {
        return;
    }

    Serial.print(F(" 📡 Connecting to MQTT... "));

    uint8_t retries = 3;
    while ((ret = mqtt.connect()) != 0) {
        Serial.println(mqtt.connectErrorString(ret));
        Serial.println(F(" 🔄 Retrying MQTT in 5 seconds..."));
        mqtt.disconnect();
        delay(5000);
        retries--;
        if (retries == 0) {
            Serial.println(F(" ❌ MQTT connection failed. System halted."));
            while (1);
        }
    }
    Serial.println(F(" ✅ MQTT Connected!\n"));
}

void updateMotor() {
    if (!motorRunning) {
        return; // Motor idle, nothing to do
    }

    // Check if spray duration completed
    if (millis() - motorStartTime >= SPRAY_DURATION) {
        stopSprayMotor();
        return;
    }

    // Sweep motor back and forth
    static int direction = 1; // 1 = forward, -1 = backward

    servoPos += direction * 2; // Move 2 degrees at a time (faster sweep)

    if (servoPos >= 180) {
        servoPos = 180;
        direction = -1; // Reverse direction
    } else if (servoPos <= 10) {
        servoPos = 10;
        direction = 1; // Reverse direction
    }

    myservo.write(servoPos);
}

```

